

Experiment 1

AIM: Develop Student Admission Management Module using OOP Concepts (Class, Object, Constructor, Access Modifiers).

Source:

A:\Btech 5\ .NET\code\pr1\pr1\Program.cs

Code:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace pr3
{
    // Student Class
    public class Student
    {
        private int number;
        private string name;
        private string course;
        private int semester;
        private double fees;

        // Constructor
        public Student(int num, string n, string c, int sem, double f)
        {
            number = num;
            name = n;
            course = c;
            semester = sem;
            fees = f;
        }

        // Scholarship Method
        public void Scholarship()
        {
            double percentage;

            Console.Write("\nEnter Student Percentage: ");
            percentage = Convert.ToDouble(Console.ReadLine());

            if (percentage >= 75)
            {
                Console.WriteLine("Congratulations! Student is eligible for Scholarship.");
            }
        }
    }
}
```

```
// 20% fee discount
fees = fees - (fees * 20 / 100);

    Console.WriteLine("Updated Fees after Scholarship: " + fees);
}
else
{
    Console.WriteLine("Student is NOT eligible for Scholarship.");
    Console.WriteLine("Fees remains: " + fees);
}
}

// Display Method
public void Display()
{
    Console.WriteLine("\n----- Student Admission Details -----");
    Console.WriteLine("Student Number : " + number);
    Console.WriteLine("Student Name : " + name);
    Console.WriteLine("Course : " + course);
    Console.WriteLine("Semester : " + semester);
    Console.WriteLine("Fees : " + fees);
}

}

// Main Class
class Program
{
    static void Main(string[] args)
    {
        int number, semester;
        string name, course;
        double fees;

        Console.Write("Enter Student Number: ");
        number = Convert.ToInt32(Console.ReadLine());

        Console.Write("Enter Student Name: ");
        name = Console.ReadLine();

        Console.Write("Enter Course: ");
        course = Console.ReadLine();

        Console.Write("Enter Semester: ");
        semester = Convert.ToInt32(Console.ReadLine());

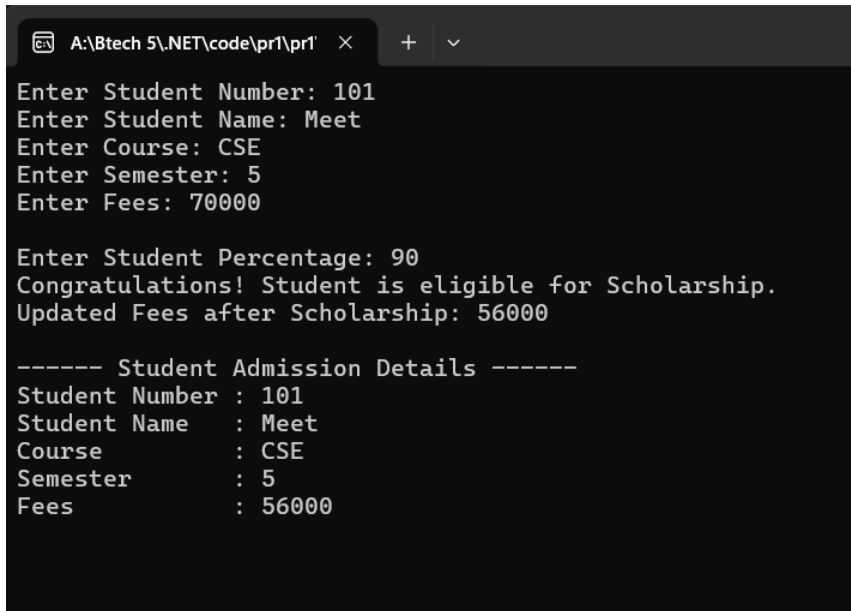
        Console.Write("Enter Fees: ");
        fees = Convert.ToDouble(Console.ReadLine());

        // Constructor
        Student s1 = new Student(number, name, course, semester, fees);

        // Scholarship Check
        s1.Scholarship();
    }
}
```

```
// Display Final Details  
s1.Display();  
  
Console.ReadKey();  
}  
}  
}
```

Output: Screenshot:



```
A:\Btech 5\NET\code\pr1\pr1  
Enter Student Number: 101  
Enter Student Name: Meet  
Enter Course: CSE  
Enter Semester: 5  
Enter Fees: 70000  
  
Enter Student Percentage: 90  
Congratulations! Student is eligible for Scholarship.  
Updated Fees after Scholarship: 56000  
  
----- Student Admission Details -----  
Student Number : 101  
Student Name   : Meet  
Course         : CSE  
Semester       : 5  
Fees           : 56000
```

GitHub URL:

<https://github.com/bosamiyadivyesh/.net-lerning/tree/test/practical1>

Experiment 2

AIM: Design Employee Payroll System using Inheritance, Interface & Polymorphism

Source:

C:\Users\ASUS\source\repos\practicals\practical-2\practical-2\

Code:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace practical_2
{
    interface IPayroll
    {
        double CalculateSalary();
    }

    abstract class Employee : IPayroll
    {
        public int EmployeeId;
        public string EmployeeName;
        public double BasicSalary;

        public Employee(int id, string name, double salary)
        {
            EmployeeId = id;
            EmployeeName = name;
            BasicSalary = salary;
        }

        public abstract double CalculateSalary();

        public void Display(string type)
        {
            Console.WriteLine("\n----- Employee Details -----");
            Console.WriteLine("Employee Type : " + type);
            Console.WriteLine("Employee ID : " + EmployeeId);
            Console.WriteLine("Employee Name : " + EmployeeName);
            //Console.WriteLine("Basic Salary : " + BasicSalary);
            Console.WriteLine("Gross Salary : " + CalculateSalary());
            Console.ReadLine();
        }
    }
}
```

```
class FullTimeEmployee : Employee
{
    public FullTimeEmployee(int id, string name, double salary)
        : base(id, name, salary)
    {
    }

    public override double CalculateSalary()
    {
        double Hra = BasicSalary * 0.10;
        double Da=BasicSalary * 0.20;
        return BasicSalary + Hra + Da;
    }

}

class PartTimeEmployee : Employee
{
    public PartTimeEmployee(int id, string name, double salary)
        : base(id, name, salary)
    {
    }

    public override double CalculateSalary()
    {
        return BasicSalary + (BasicSalary * 0.10);
    }
}

class Program
{
    static void Main(string[] args)
    {
        Console.WriteLine("=====");
        Console.WriteLine(" Employee Payroll Management");
        Console.WriteLine("=====");

        Console.WriteLine("\nSelect Employee Type");
        Console.WriteLine("1. Full Time Employee");
        Console.WriteLine("2. Part Time Employee");

        Console.Write("\nEnter Choice : ");
        int choice = Convert.ToInt32(Console.ReadLine());

        Employee emp;
        if (choice == 1)
        {

            Console.Write("\nEnter Employee ID : ");
            int id = Convert.ToInt32(Console.ReadLine());
```

```
        Console.Write("Enter Employee Name : ");
        string name = Console.ReadLine();

        Console.Write("Enter Basic Salary : ");
        double salary = Convert.ToDouble(Console.ReadLine());

        emp = new FullTimeEmployee(id, name, salary);
        emp.Display("Full Time");

    }
    else if (choice == 2) {

        Console.Write("\nEnter Employee ID : ");
        int id = Convert.ToInt32(Console.ReadLine());

        Console.Write("Enter Employee Name : ");
        string name = Console.ReadLine();

        Console.Write("Enter Basic Salary : ");
        double salary = Convert.ToDouble(Console.ReadLine());

        emp = new PartTimeEmployee(id, name, salary);
        emp.Display("Part Time");
    }
    else
    {
        Console.WriteLine("Invalid Choice.");
    }

    Console.ReadKey();
}

}
```

Output: Screenshot:

```
C:\Users\ASUS\source\repos\ x + v

=====
Employee Payroll Management
=====

Select Employee Type
1. Full Time Employee
2. Part Time Employee

Enter Choice : 1

Enter Employee ID : 47
Enter Employee Name : divyesh
Enter Basic Salary : 75000

----- Employee Details -----
Employee Type : Full Time
Employee ID : 47
Employee Name : divyesh
Gross Salary : 97500
```

```
C:\Users\ASUS\source\repos\ x + v

=====
Employee Payroll Management
=====

Select Employee Type
1. Full Time Employee
2. Part Time Employee

Enter Choice : 2

Enter Employee ID : 0047
Enter Employee Name : divyesh
Enter Basic Salary : 25000

----- Employee Details -----
Employee Type : Part Time
Employee ID : 47
Employee Name : divyesh
Gross Salary : 27500
```

GitHub URL:

<https://github.com/bosamiyadivvish/.net-lerning/tree/test/practcal-2>

Experiment 3

AIM: Develop Expense Tracking Module with Exception Handling.

Source:

A:\Btech 5\ .NET\code\pr3\pr3\Program.cs

Code:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace prctical_3
{
    class InvalidAmountException : Exception
    {
        public InvalidAmountException(string message) : base(message)
        { }
    }
    internal class Program
    {
        static void Main(string[] args)
        {
            string[] category = new string[10];
            double[] amount = new double[10];
            int count = 0;
            int choice;

            do
            {
                Console.WriteLine("\n===== Expense Tracker =====");
                Console.WriteLine("1. Add Expense");
                Console.WriteLine("2. Display Expenses");
                Console.WriteLine("3. Exit");
                Console.Write("Enter your choice: ");

                try
                {
                    choice = Convert.ToInt32(Console.ReadLine());

                    switch (choice)
                    {
                        case 1:
                            if (count == 10)
                            {
                                Console.WriteLine("Expense storage is full.");
                                break;
                            }

```

```
try
{
    Console.Write("Enter Category: ");
    category[count] = Console.ReadLine();

    foreach (char ch in category[count])
    {
        if (!char.IsLetter(ch) && ch != ' ')
        {
            throw new Exception("Category should contain only
letters.");
        }
    }
}
catch (Exception ex)
{
    Console.WriteLine(ex.Message);
    continue;    // Return to the menu
}

Console.Write("Enter Amount: ");

try
{
    amount[count] = Convert.ToDouble(Console.ReadLine());

    if (amount[count] <= 0)
    {
        throw new InvalidAmountException("Amount must be
greater than 0.");
    }

    count++;
    Console.WriteLine("Expense Added Successfully.");
}
catch (InvalidAmountException ex)
{
    Console.WriteLine(ex.Message);
}

break;

case 2:
    if (count == 0)
    {
        Console.WriteLine("No Expenses Found.");
    }
    else
    {
        double total = 0;

        Console.WriteLine("\nExpense List");
        Console.WriteLine("-----");

        for (int i = 0; i < count; i++)
```

```
        {
            Console.WriteLine("Category : " + category[i]);
            Console.WriteLine("Amount : " + amount[i]);
            Console.WriteLine("-----");

            total += amount[i];
        }

        Console.WriteLine("Total Expense = " + total);
    }
    break;

    case 3:
        Console.WriteLine("Thank You!");
        break;

    default:
        Console.WriteLine("Invalid Choice.");
        break;
    }
}
catch (FormatException)
{
    Console.WriteLine("Please enter valid input.");
    choice = 0;
}

} while (choice != 3);

}

}
```

```
Microsoft Visual Studio Debug x + v
2. Display Expenses
3. Exit
Enter your choice: 1
Enter Category: food
Enter Amount: 15000
Expense Added Successfully.

===== Expense Tracker =====
1. Add Expense
2. Display Expenses
3. Exit
Enter your choice: 2

Expense List
-----
Category : food
Amount   : 15000
-----
Total Expense = 15000

===== Expense Tracker =====
1. Add Expense
2. Display Expenses
3. Exit
Enter your choice: 3
Thank You!

C:\Users\ASUS\source\repos\practicals\practical-3
0 (0x0).
Press any key to close this window . . .
```

```
C:\Users\ASUS\source\repos\ x + v

===== Expense Tracker =====
1. Add Expense
2. Display Expenses
3. Exit
Enter your choice: 1
Enter Category: food
Enter Amount: 15000
Expense Added Successfully.

===== Expense Tracker =====
1. Add Expense
2. Display Expenses
3. Exit
Enter your choice:
```

GitHub URL:

<https://github.com/bosamiyadivyesh/.net-lerning/tree/test/practical-3>

Experiment 4

AIM: Build Online Event Registration Portal using ASP.NET Controls & Validation

Registartion.aspx:-

[illegible]

```

        <asp:TableCell><asp:TextBox      ID="email"      runat="server"
type="email" AutoPostBack="true" ></asp:TextBox>
        <asp:Label      ID="uemail"      runat="server"
ForeColor="Red"></asp:Label>
    </asp:TableCell>
</asp:TableRow>
<asp:TableRow>
    <asp:TableCell>Mobile Number:</asp:TableCell>
    <asp:TableCell><asp:TextBox      ID="mbnumber"      runat="server"
type="number" AutoPostBack="true" ></asp:TextBox>
        <asp:Label      ID="umbnumber"      runat="server"
ForeColor="Red"></asp:Label>
    </asp:TableCell>
</asp:TableRow>
<asp:TableRow>
    <asp:TableCell>Collage:</asp:TableCell>
    <asp:TableCell><asp:TextBox      ID="collage"      runat="server"
type="text" AutoPostBack="true" ></asp:TextBox>
        <asp:Label      ID="ucollage"      runat="server"
ForeColor="Red"></asp:Label>
    </asp:TableCell>
</asp:TableRow>
<asp:TableRow>
    <asp:TableCell>Department:</asp:TableCell>
    <asp:TableCell><asp:RadioButtonList      ID="department"
runat="server" AutoPostBack="true" >
        <asp:ListItem Value="cse">CSE</asp:ListItem>
        <asp:ListItem Value="mce">Mechanical</asp:ListItem>
        <asp:ListItem Value="civil">Civil</asp:ListItem>
        <asp:ListItem Value="chemical">Chemical</asp:ListItem>
    </asp:RadioButtonList>
        <asp:Label      ID="udepartment"      runat="server"
ForeColor="Red"></asp:Label>
    </asp:TableCell>

```

```

    </asp:TableRow>
    <asp:TableRow>
        <asp:TableCell>Event:</asp:TableCell>
        <asp:TableCell>
            <asp:DropDownList          ID="events"          runat="server"
AutoPostBack="true">
                <asp:ListItem Value="">Select presentation</asp:ListItem>
                <asp:ListItem      Value="paper      presentation">Paper
presentation</asp:ListItem>
                <asp:ListItem      Value="post      presentation">Poster
presentation</asp:ListItem>
                <asp:ListItem Value="quiz">Quiz</asp:ListItem>
                <asp:ListItem Value="fung">Fun Games</asp:ListItem>
            </asp:DropDownList>
            <asp:Label          ID="uevent"          runat="server"
ForeColor="Red"></asp:Label>

        </asp:TableCell>
    </asp:TableRow>

    <asp:TableRow>
        <asp:TableCell>Gender :</asp:TableCell>
        <asp:TableCell>
            <asp:RadioButton ID="male" runat="server" value="male"
GroupName="gender" AutoPostBack="true" />Male
            <asp:RadioButton ID="female" runat="server" value="female"
GroupName="gender" AutoPostBack="true" />Female

            <asp:Label          ID="ugender"          runat="server"
ForeColor="Red"></asp:Label>
        </asp:TableCell>
    </asp:TableRow>
  
```

```

        <asp:TableRow>
        <asp:TableCell>Skills :</asp:TableCell>
        <asp:TableCell>
            <asp:CheckBoxList ID="skills" runat="server" AutoPostBack="true" >
                <asp:ListItem Value="c#">c#</asp:ListItem>
                <asp:ListItem Value="java">Java</asp:ListItem>
                <asp:ListItem Value="python">Python</asp:ListItem>
                <asp:ListItem Value="ai">Ai</asp:ListItem>
            </asp:CheckBoxList>
            <asp:Label
                ID="uskills"
                runat="server"
                ForeColor="Red"></asp:Label>

        </asp:TableCell>
    </asp:TableRow>

    <asp:TableRow>

        <asp:TableCell>Address:</asp:TableCell>
        <asp:TableCell>
            <textarea id="Address" name="address"></textarea>
            <asp:Label
                ID="uaddress"
                runat="server"
                ForeColor="Red"></asp:Label>
        </asp:TableCell>
    </asp:TableRow>

    <asp:TableFooterRow>
        <asp:TableCell>
            <asp:CheckBox ID="term" runat="server"/>term and condition<br
    />
            <asp:Label
                ID="terms"
                runat="server"
                ForeColor="red"></asp:Label>
            <br />
        </asp:TableCell>
    </asp:TableFooterRow>
    </asp:Table>
    </div>
    </script>
</body>
</html>

```

```
<asp:Button ID="btn1" runat="server" Text="Registartion"
OnClick="btn_Click"/>
</asp:TableCell>
</asp:TableFooterRow>
</asp:Table>
```

```
</form>
```

```
</body>
```

```
</html>
```

Registrstion.aspx.cs:-

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text.RegularExpressions;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace practical4
{
    public partial class Registartion : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
            if (IsPostBack)
            {
                uname.Text = "";
                uemail.Text = "";
                umbnumber.Text = "";
            }
        }
    }
}
```

```
ucollage.Text = "";
udepartment.Text = "";
uevent.Text = "";
ugender.Text = "";
uskills.Text = "";
uaddress.Text = "";

// Name
if (string.IsNullOrEmpty(username.Text))
    uname.Text = "Please enter your name.";

// Email
if (string.IsNullOrEmpty(email.Text))
    uemail.Text = "Please enter your email.";
else if (!Regex.IsMatch(email.Text,
@"^[^@\s]+@[\s]+\.[^\s]+$"))
    uemail.Text = "Invalid email.";

// Mobile
if (string.IsNullOrEmpty(mnumber.Text))
    umbnumber.Text = "Please enter mobile number.";
else if (mnumber.Text.Length != 10)
    umbnumber.Text = "Mobile number must be 10 digits.";

// College
if (string.IsNullOrEmpty(collage.Text))
    ucollage.Text = "Please enter college.";

// Department
if (department.SelectedIndex == -1)
    udepartment.Text = "Please select department.";

// Event
if (events.SelectedIndex == -1)
```

```
uevent.Text = "Please select event.";

// Gender
if (!male.Checked && !female.Checked)
    ugender.Text = "Please select gender.";

// Skills
bool selected = false;
foreach (ListItem item in skills.Items)
{
    if (item.Selected)
    {
        selected = true;
        break;
    }
}

if (!selected)
    uskills.Text = "Please select at least one skill.";

// Address
if (string.IsNullOrEmpty(Request.Form["address"]))
    uaddress.Text = "Please enter address.";
}

}

protected void btn_Click(object sender, EventArgs e)
{
    if (!term.Checked)
    {
        terms.Text = "Please checked Terms and Condition";
    }
}
```

```
        return;
    }
    string gender = "";

    if (male.Checked)
        gender = "Male";
    else if (female.Checked)
        gender = "Female";

    string skill = "";

    foreach (ListItem item in skills.Items)
    {
        if (item.Selected)
        {
            skill += item.Text + " ";
        }
    }

    string address = Request.Form["address"];

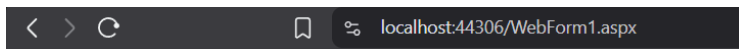
    Response.Write("<h2>Registration Details</h2>");
    Response.Write("<b>Name :</b> " + username.Text + "<br/>");
    Response.Write("<b>Email :</b> " + email.Text + "<br/>");
    Response.Write("<b>Mobile Number :</b> " + mbnumber.Text +
"<br/>");
    Response.Write("<b>College :</b> " + collage.Text + "<br/>");
    //      Response.Write("<b>Department      :</b>      "      +
department.SelectedItem.Text + "<br/>");
    Response.Write("<b>Event :</b> " + events.SelectedItem.Text +
"<br/>");
    Response.Write("<b>Gender :</b> " + gender + "<br/>");
    Response.Write("<b>Skills :</b> " + skill + "<br/>");
```

```
Response.Write("<b>Address :</b> " + address + "<br/>");
```

```
}
```

```
}
```

```
}
```



Form Registration

Name:		Please enter your name.
Email:		Please enter your email.
Mobile Number:		Please enter mobile number.
Collage:		Please enter college.
Department:	☐ CSE ☐ Mechanical ☐ Civil ☐ Chemical	Please select department.
Event:	Select presentation	
Gender :	☐ Male ☐ Female	Please select gender.
Skills :	☐ c# ☐ Java ☐ Python ☐ Ai	Please select at least one skill.
Address:		Please enter address.
☐ term and condition		
Registration		

< > ↺  localhost:44306/WebForm1.aspx

Form Registration

Name:	Divyesh
Email:	divyes@gmail.com
Mobile Number:	1234567890
Collage:	MU
Department:	☒ CSE ☐ Mechanical ☐ Civil ☐ Chemical
Event:	Fun Games
Gender :	☒ Male ☐ Female
Skills :	☐ c# ☒ Java ☐ Python ☐ Ai
Address:	Rajkot
☒ term and condition Registartion	

< > ↺   localhost:44306/WebForm1.aspx

Registration Details

Name : Divyesh
Email : divyes@gmail.com
Mobile Number : 1234567890
College : MU
Event : Fun Games
Gender : Male
Skills : Java
Address : Rajkot

Form Registration

Name:	
Email:	
Mobile Number:	
Collage:	
Department:	☐ CSE ☐ Mechanical ☐ Civil ☐ Chemical
Event:	Select presentation
Gender :	☐ Male ☐ Female
Skills :	☐ c# ☐ Java ☐ Python ☐ Ai
Address:	
☒ term and condition Registartion	

GitHub URL:

<https://github.com/bosamiyadivyes/.net-lerning/tree/test/practical4>